

DBMS – REVIEW AND RATING MANAGEMENT

Project Title: Gaming-Accessories E-Commerce Order Management System

1. Database Design

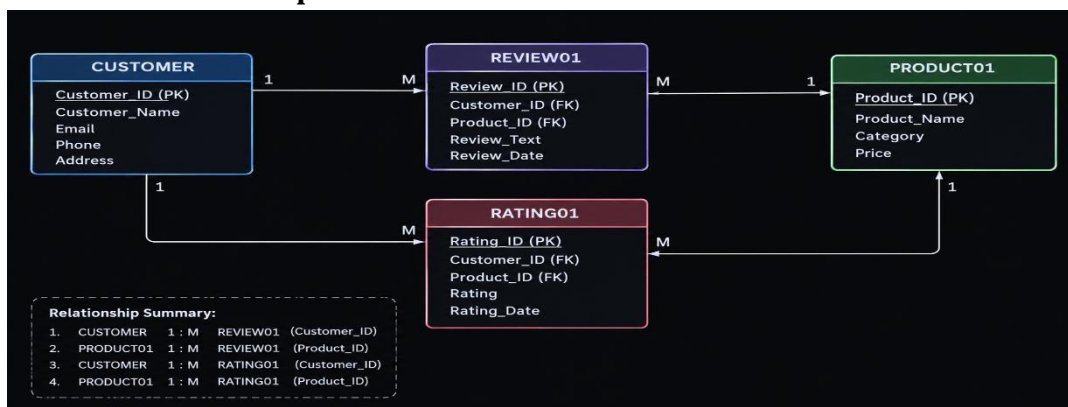
1.1 Review Table

Attribute	Key	Description
Inventory_ID	Primary Key (PK)	Unique inventory ID
Product_ID	Foreign Key (FK)	ID of the product
Quantity_Available	-	Available quantity in stock
Last_Updated	-	Date of last stock update

1.2 Rating Table

Attribute	Key	Description
Rating_ID	Primary Key (PK)	Unique rating ID
Product_ID	Foreign Key (FK)	ID of the product being rated
Customer_ID	Foreign Key (FK)	ID of the customer giving the rating
Rating_Value	-	Rating given by the customer (1–5)
Rating_Date	-	Date when the rating was given
Comments	-	Customer's comments about the product

2. Table Relationships



3. Table Creation

3.1 Create Review Table

```
SQL> CREATE TABLE review01 (  
2     review_id NUMBER PRIMARY KEY,  
3     product_id NUMBER,  
4     customer_name VARCHAR2(50),  
5     rating NUMBER(1),  
6     review_text VARCHAR2(200),  
7     review_date DATE,  
8     FOREIGN KEY (product_id) REFERENCES product01(product_id)  
9 );
```

Table created.

3.2 Create Rating Table

```
SQL> CREATE TABLE rating01 (  
2     rating_id NUMBER PRIMARY KEY,  
3     product_id NUMBER,  
4     customer_name VARCHAR2(50),  
5     rating NUMBER(1),  
6     rating_date DATE,  
7     FOREIGN KEY (product_id) REFERENCES product01(product_id)  
8 );
```

Table created.

4. Review Data Insertion

```
SQL> INSERT INTO review01  
2 VALUES (1, 101, 'Arun', 5, 'Excellent laptop', SYSDATE);  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO review01  
2 VALUES (2, 102, 'Priya', 4, 'Good smartphone', SYSDATE);  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO review01  
2 VALUES (3, 103, 'Rahul', 5, 'Very good sound quality', SYSDATE);  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO review01  
2 VALUES (4, 104, 'Anu', 3, 'Average product', SYSDATE);  
  
1 row created.  
  
SQL>  
SQL> INSERT INTO review01  
2 VALUES (5, 105, 'Kavin', 4, 'Good keyboard', SYSDATE);  
  
1 row created.
```

5. Rating Data Insertion

```
SQL> INSERT INTO rating01
  2 VALUES (1, 101, 'Arun', 5, SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO rating01
  2 VALUES (2, 102, 'Priya', 4, SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO rating01
  2 VALUES (3, 103, 'Rahul', 5, SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO rating01
  2 VALUES (4, 104, 'Anu', 3, SYSDATE);

1 row created.

SQL>
SQL> INSERT INTO rating01
  2 VALUES (5, 105, 'Kavin', 4, SYSDATE);

1 row created.
```

6. Retrieve Product Review Details

```
SQL> SELECT p.product_id, p.product_name,
  2         r.customer_name, r.rating,
  3         r.review_text, r.review_date
  4 FROM product01 p
  5 JOIN review01 r
  6 ON p.product_id = r.product_id;
```

```
PRODUCT_ID
-----
PRODUCT_NAME
-----
CUSTOMER_NAME                                RATING
-----
REVIEW_TEXT
-----
REVIEW_DA
-----
          101
HP Laptop
Arun                                           5
```

7. Calculate Average Product Ratings

```
SQL> SELECT p.product_id,
2         p.product_name,
3         AVG(r.rating) AS average_rating
4 FROM product01 p
5 JOIN review01 r
6 ON p.product_id = r.product_id
7 GROUP BY p.product_id, p.product_name;
```

PRODUCT_ID	PRODUCT_NAME	AVERAGE_RATING
101	HP Laptop	5
102	Dell Laptop	4

8. Identify Highly Rated Products

Products with an average rating of 4 or above can be displayed using HAVING.

```
SQL> SELECT Product_ID,
2         AVG(Rating) AS Average_Rating
3 FROM rating01
4 GROUP BY Product_ID
5 HAVING AVG(Rating) >= 4
6 ORDER BY Average_Rating DESC;
```

PRODUCT_ID	AVERAGE_RATING
103	5
101	5
105	4
102	4

9. Product Rating Report

A combined report can be generated using the Rating and Product tables.

```
SQL> SELECT Product_ID,
2         COUNT(Rating_ID) AS Total_Ratings,
3         AVG(Rating) AS Average_Rating,
4         MAX(Rating) AS Highest_Rating,
5         MIN(Rating) AS Lowest_Rating
6 FROM rating01
7 GROUP BY Product_ID
8 ORDER BY Average_Rating DESC;
```

PRODUCT_ID	TOTAL_RATINGS	AVERAGE_RATING	HIGHEST_RATING	LOWEST_RATING
101	1	5	5	5
103	1	5	5	5
105	1	4	4	4
102	1	4	4	4
104	1	3	3	3